Although the reasoning here is straightforward, the conclusion may nevertheless be confusing. Since the galaxies are all rushing apart as space expands, clocks that move along with one or another galaxy are also rushing apart. What's more, they're moving relative to each other at an enormous variety of speeds determined by the enormous variety of distances between them. Won't this motion cause the clocks to fall out of synchronization, as Einstein taught us with special relativity? For a number of reasons, the answer is no; here is one particularly useful way to think about it.

Recall from Chapter 3 that Einstein discovered that clocks that move *through* space in different ways tick off time at different rates (because they divert different amounts of their motion through time into motion through space; remember the analogy with Bart on his skateboard, first heading north and then diverting some of his motion to the east). But the clocks we are now discussing are *not* moving through space at all. Just as each penny is glued to one point on the balloon and only moves relative to other pennies because of the swelling of the balloon's surface, each galaxy occupies one region of space and, for the most part, only moves relative to other galaxies because of the expansion of space. And this means that, with respect to space itself, all the clocks are actually stationary, so they tick off time identically. It is precisely these clocks—*clocks whose only motion comes from the expansion of space*—that provide the synchronized cosmic clocks used to measure the age of the universe.

